

# ITAS170 — Lab 4

## Local models versus frontier models — with Claude as the yardstick

**Time:** one class, Monday 9:00–2:00 · **Marks:** 20

**Work from a terminal, and keep the session log.** Today one of the models you log is running on the machine in front of you, and you can watch it work.

Work in `~/code/ITAS170/week04/`.

**Get the starter files first:** download `itas170-week04-student.zip` from the week 4 section on Moodle and unzip it into `~/code/ITAS170/week04/`. `ls` should show `prompts.md`, `compare-template.md`, `scoreboard-template.csv`, `sizing-worksheet.md`, `claims.md`, `local-vs-hosted.md` and `README.txt`. Then check the models are there:

```
ollama list
```

You should see `qwen3:4b`, `qwen3:8b` and `hf.co/Qwen/Qwen3-4B-GGUF:Q4_K_M`. They are pulled once per desktop before class, and a model pulled on a desktop is there for every account on it. If the list is short, pull **in this order** — the 4B first, because Part 1 needs it — and read on while it downloads (a 2.5 GB pull took under five minutes here on a busy day):

```
ollama pull qwen3:4b                # 2.5 GB — Part 1 needs it
ollama pull qwen3:8b                # 5.2 GB — second terminal
ollama pull hf.co/Qwen/Qwen3-4B-GGUF:Q4_K_M  # 2.5 GB — Part 2
```

## How the day runs

| Time        |                              |                                                                                         |
|-------------|------------------------------|-----------------------------------------------------------------------------------------|
| 9:00–9:40   | <b>Intro A</b>               | What "local" means; quantization; tokens per second; the frontier                       |
| 9:40–10:30  | <b>Session A — Parts 1–2</b> | ~60 min hands-on — Part 2 runs into the break if it must                                |
| 10:30–11:00 | Break                        |                                                                                         |
| 11:00–11:40 | <b>Intro B</b>               | How to compare three models fairly; read before you sign; AGI, RSI, and reading a claim |

| Time            |                              |                                                   |
|-----------------|------------------------------|---------------------------------------------------|
| 11:40–<br>12:30 | <b>Session B — Parts 3–5</b> | Instructor breaks for lunch around noon           |
| 12:30–<br>2:00  | <b>Finish + demos</b>        | Help, then <b>show+tell marking</b> as you finish |

**You are marked in person, and demos run all day. 6 of the 20 marks are for explaining your work** — this week's opening question is "your model fits in RAM; why is it slow?"

**Three models today.** The **local** ones are qwen3:4b and qwen3:8b: weights on this disk, running on this CPU — the desktops in 180-332 have 64 GB, so both fit, and the difference between them is Route A — nothing leaving the room. The **yardstick** is **Claude Opus 5.5**, through your Team seat: open the Claude app, pick Opus 5.5 in the model picker (check it; do not assume the default) and leave it there for the rest of the term. Every comparison this course makes is against this one model, so the rows stay comparable from week to week. The third is **your assigned model**: one of three Chinese labs' flagships, used as a hosted web chat — **Kimi K3** (kimi.com; if the site now says K3.1, write that), **DeepSeek V4.1 Flash** (chat.deepseek.com, which has served it since 10 September) or **GLM-5.3** (chat.z.ai — pick it in the model menu; the default there is the smaller GLM-5.3-Flash). The instructor assigns you one at 11:00; write the name the site shows, not the one on this page, at the top of `compare.md`. You do not create that account until you have read what you are agreeing to (Part 3, step 0), and **nobody has to sign up at all** — route P is there for exactly that: your instructor runs the prompts for you. Name every model in your files; which ones you used is data, not a secret.

## Part 1 — Run it, and watch it · 25 min

Open two terminals side by side. In the first:

```
cd ~/code/ITAS170/week04
ollama run --verbose --think=false qwen3:4b "Explain exactly what this shell command does,
option by option, in at most five lines. Do not suggest alternatives.  sort -k10,10n
access.log | tail -1"
```

While it is answering, in the second terminal:

```
ollama ps
```

| NAME     | ID           | SIZE   | PROCESSOR | CONTEXT | UNTIL              |
|----------|--------------|--------|-----------|---------|--------------------|
| qwen3:4b | 359d7dd4bcda | 3.2 GB | 100% CPU  | 4096    | 4 minutes from now |

Three facts in one line: how much memory the model is using, **what is running it** (100% CPU: the lab desktops have an Intel UHD 770 on the processor (could vary across workstations) and no graphics card, and Ollama does not use the UHD, so this is what you will see), and the size of its desk — 4096 tokens, the default, whatever the model could do.

When the answer finishes, `--verbose` prints a block like this. It is from the machine the handout was written on; **on the lab desktops the 4B came out at about 15 tokens/s** when one was measured on 28 September, and the 8B should be about half that. Your numbers are the point:

```
total duration:      21.366925459s
load duration:       4.096804541s
prompt eval count:   50 token(s)
prompt eval rate:    118.21 tokens/s
eval count:          339 token(s)
eval rate:           20.12 tokens/s
```

**eval rate is the number.** It is how fast the model generates — tokens per second — and it is what you will compare all day. `load duration` is the cost of getting the weights into memory (large the first time, tiny while the model stays loaded). `prompt eval` is reading your question; `eval` is writing the answer. The very first run on a desktop can report a `total duration` of minutes: that is the weights coming off the disk cold, plus the download if the pull was still finishing. It is not the eval rate. Run it again and read the second `load duration`.

Run the same prompt **two more times** so you have three eval rates. Then run it once **without** `--think=false` and watch the model reason before it answers — and what that does to the time.

Write `local.md`: the `ollama ps` line, your three eval rates and their median, the thinking-on time, and two sentences: what the `load duration` did between run one and run two, and what `--think` bought or cost.

**Hand in:** `local.md`.

## Part 2 — Size it · 35 min

Copy `sizing-worksheet.md` to `sizing.md`. The rule of thumb is on it: **memory  $\approx$  parameters  $\times$  bytes per parameter + working room**, and Ollama's default download is a **Q4** quantization at about 0.6 bytes per parameter.

Get the facts for the two models you have:

```
ollama show qwen3:4b
ollama show qwen3:8b
```

Read `parameters`, `quantization` and `context length`. Now the same model a second way — **pulled straight from its own page on Hugging Face**, the hub where open models are published, instead of from Ollama's library:

```
ollama show hf.co/Qwen/Qwen3-4B-GGUF:Q4_K_M
ollama list
```

Same family, same size, same quantization — and not the same file. Compare the two `show` outputs and the two IDs in `ollama list`, and write one line in `sizing.md`: what differs, and how you would know which weights you are actually running. Then open that model's page ([huggingface.co/Qwen/Qwen3-4B-GGUF](https://huggingface.co/Qwen/Qwen3-4B-GGUF) — the "model card") and read three things: its **licence**, what it says about **training data**, and its **limitations** section. Put one sentence about each in `sizing.md`. Fill in the table — every row — **predicting first**. Then check one prediction against reality: run `qwen3:8b` (or `4b`) and read `SIZE` in `ollama ps`. Write the difference down.

The last line of the worksheet is the demo question. **A model that fits in system RAM runs** — you just proved it — **but at what speed?** Your Part 1 eval rate is the evidence. Say what would change it.

**Hand in:** `sizing.md` — the table, the hub-versus-library line, the three model-card sentences, the fits-but-slow sentence.

## Part 3 — The blind comparison · 55 min

Copy `compare-template.md` to `compare.md`.

**Step 0 — read before you sign (10 min).** Open your assigned model's privacy policy — the addresses are in `local-vs-hosted.md`, under The three policies — and answer the template's five questions from the page itself: who controls the data and in which country, where it is stored, whether your text is used to train the model and how to opt out, how you delete, and your decision — and the policy's **date**: the page you get can depend on where you sign up from, and one of these addresses has served two different versions this term.

Then, and only then, create the account with an email you are willing to give — or choose **route P**: ask your instructor to run the prompts for you with their Kimi or DeepSeek account. No reason needed, no account, no payment card. You still do step 0 — reading a policy needs no account, and your decision line says route P. The answers come back as a file with each prompt's wall time and the model ID the service reported; you check and score them exactly as everyone else does, and you name that model in your table and your scoreboard rows. Kimi or DeepSeek, whichever you are given — if you were assigned GLM-5.3, write down which one ran yours. The marks are for the checking, not the account. Nothing private goes into any of these windows today; the six prompts are public text.

`prompts.md` has **six prompts**: explain a command, read a log, write a command, find a bug, a recent fact, and a question about history that some services will not answer. For each prompt:

1. Paste it to the local model (`ollama run --think=false qwen3:4b`; the 8b fits too and runs at about half the speed — name whichever you used) in a **fresh** run, to **Opus 5.5** in a **fresh** conversation, and to your assigned model in a fresh conversation.
2. Time all three (a phone timer is fine; for the local one, `total duration`).
3. **Check the answer yourself before scoring it** — you have the log lines for P2 (count them), you can run P3 and P4, and P5 and P6 have right answers the instructor will confirm at the demo.
4. Score each 0-2 on **correct, complete, checkable**.

**About P6.** It asks about the events of June 1989 in Beijing, which the government of the country where three of these models are made restricts. It is not a trick and not a test of your opinions: the result is whether each model **answered, refused, or changed the subject** — write down its exact words. Notice what the local column tells you. The weights on your desktop have no service in front of them, so whatever they do on P6 came with the training, not from a filter on a website. Score it like the others; a refusal scores 0 on complete.

The scores are yours to defend. At the bottom, four short answers: where local won and why, where the yardstick won and why, where your assigned model beat the yardstick or lost to it and on what, and which you would hand a private file and which a hard question — read `local-vs-hosted.md` first: the answer is not only about capability, and the table names the privacy, security and ethics trade-offs you are choosing between.

**The scoreboard starts here.** Copy `scoreboard-template.csv` to `~/code/ITAS170/scoreboard.csv` — once; it lives at the root of your repository and grows one week at a time from now until Launch Day. This week's task is **P4**: add three rows, one per model, in the template's columns. Same question, three models, every Monday: by December you will have your own answer to "which of these is actually good at what".

**Hand in:** `compare.md`, and `scoreboard.csv` with this week's three rows.

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## Part 4 — Read a claim like a traceback · 25 min

Open `claims.md`. Three texts: an announcement about how good a model is, a launch post in which one of this afternoon's Chinese labs compares its model with Claude, and a security report accusing rivals of distilling a model — training a cheaper model on a frontier model's answers. For each, **five lines** — what is claimed, who measured it, what you would need to know to check, how it could be true and still misleading, your verdict.

Then the two words from the morning: in your own words, what **AGI** means, what **RSI** means, and which of the two claims has anything to do with either.

**Hand in:** `claims.md`, filled in.

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## Part 5 — Make it yours · 25 min

**Three routes. Each is worth the full four marks.**

**Route A — a second local model.** Run the six prompts through `qwen3:8b` as well (or another model of a different family you pull — ask first; the network is shared). Add a column to `compare.md`. Does twice the parameters buy accuracy, and on which prompts? What did it cost in eval rate?

**Route B — a task from your own life.** Something you actually need — a message, a plan, a bit of homework from another course, a recipe scaled for six — through both models, scored the same three ways, with a paragraph on which you would trust and why.

**Route C — the desk, measured.** P2 with its twelve lines, then P2 with 120 lines (copy the block ten times). Compare `prompt eval count`, `prompt eval duration`, and whether the answer is still right. Then try `--think on` and `off` for P4. Two paragraphs on what context length and thinking cost, in seconds and in correctness.

Whichever route: commit and push.

```
cd ~/code/ITAS170
git add -A
git commit -m "Week 4: local vs frontier"
git push
```

**Hand in:** the route's additions, in the repo, pushed.

## Marking — 20 marks

| Component |                                                                                                                                                | Marks     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| C1        | Run it — ollama ps, three eval rates and the median, load and thinking explained                                                               | 2         |
| C2        | Size it — the table predicted then checked, the hub pull compared, the model card read, the fits-but-slow sentence                             | 3         |
| C3        | The blind comparison — the policy read before signing, six prompts to three models, checked before scored, four answers, three scoreboard rows | 3         |
| C4        | Read a claim — five lines for each of three claims, AGI and RSI in your words                                                                  | 2         |
| C5        | <b>Make it yours</b> — a second model, your own task, or the desk measured                                                                     | 4         |
| <b>DU</b> | <b>Demonstrates understanding</b> — marked in person                                                                                           | <b>6</b>  |
|           |                                                                                                                                                | <b>20</b> |

The **make it yours** criterion is worth **4 marks** — more than any single required part. DU is awarded from your show+tell demo, not from your files. Opening question: "your model fits in RAM; why is it slow?" Follow-up: "what was P2's right answer, and did either model get it?"

## Submitting

Your repository is the submission. For the record, also zip `~/code/ITAS170/week04/` and upload it to the week 4 section on Moodle by **Sunday 11 October** — the Sunday before Thanksgiving; there is no class on the 12th.

## On using AI for this lab

Two assistants are the subject today, so use both freely — and log both in `LOG.md`, the local one included. Hands-only moments: **you check each answer before you score it** (an assistant grading itself is not a comparison), **your predictions in Part 2 come before you ask anything**, and **the claims analysis is yours**. Name the frontier model you used in `compare.md`. Nothing private goes to it today — that is the local model's job, and part of the comparison.

## If something goes wrong

| What you see                                                                             | What it means                                                                                                                                                                                    | What to do                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ollama: command not found</code>                                                   | Not on a lab desktop — Ollama 0.34.0 is on the image (checked 28 Sep). On your own laptop: not installed.                                                                                        | Tell the instructor. Meanwhile, pair up: one machine, two students, take turns.                                                                                                                                                                                                                         |
| <code>Error: could not connect to ollama server</code>                                   | On the lab desktops Ollama is a <b>system service</b> ; <code>systemctl is-active ollama</code> should print <code>active</code> .                                                               | If it prints anything else, tell the instructor — restarting it needs <code>sudo</code> . Do <b>not</b> run <code>ollama serve</code> yourself on a desktop: that starts a second server with an empty model list. On your own laptop, <code>ollama serve</code> in a spare terminal is the right move. |
| <code>ollama list</code> is empty, or <code>pulling manifest</code> sits at 0%           | Nothing has been pulled on this desktop yet, or the lab network is shared by sixteen downloads. Models live with the service: one pull per machine serves every account on it.                   | Pull the 4B first and start Part 1 when it lands; the other two in a second terminal. A 2.5 GB pull took under five minutes here on a busy class day. Start Part 2's reading meanwhile.                                                                                                                 |
| <code>ollama ps</code> shows nothing                                                     | The model unloaded — it stays in memory about five minutes after the last request.                                                                                                               | Run a prompt and look again while it answers.                                                                                                                                                                                                                                                           |
| The answer goes on for pages when you asked for five lines                               | Small models follow instructions loosely, and thinking mode adds a reasoning pass.                                                                                                               | That is a result: score complete accordingly. <code>--think=false</code> for the timing runs.                                                                                                                                                                                                           |
| <code>SIZE</code> in <code>ollama ps</code> is bigger than your worksheet said           | Working room (the context cache) and overhead sit on top of the weights.                                                                                                                         | Expected — the worksheet's "+ working room" is exactly this. Write the difference down; it is the mark.                                                                                                                                                                                                 |
| <code>eval rate</code> is well under 10 tokens/s for the 4B (the desktops give about 15) | Another model is still loaded, the machine is busy, or you are reading the first run's <code>total duration</code> (minutes: the cold load and the download) instead of <code>eval rate</code> . | <code>free -h</code> and <code>ollama ps</code> — is something else loaded? <code>ollama stop qwen3:8b</code> . Run it again and read <code>eval rate</code> on the second run.                                                                                                                         |
| Your frontier assistant refuses P4 or P5 or asks a clarifying question                   | Assistants vary.                                                                                                                                                                                 | Answer the clarification once; score what you get. A refusal scores 0 on complete and you note why.                                                                                                                                                                                                     |
| P5's answer is confident and you cannot tell if it is right                              | That is P5's whole point.                                                                                                                                                                        | Score checkable on whether it gave you anything to check, and confirm the fact at the demo.                                                                                                                                                                                                             |



| What you see                                                                                                 | What it means                                                                                                                      | What to do                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A model refuses P6, or answers about the square's architecture instead                                       | That is the result P6 measures. Some models were trained to avoid the topic; some services filter it as well.                      | Write down its exact words, score complete 0, and say in your closing answers whether the local model did the same — that tells you whether it is the weights or the website.        |
| Opus 5.5 is not in the model picker                                                                          | The seat has a different model list than expected.                                                                                 | Tell the instructor (it is a seat setting), use the default model today, and write the model's actual name in every row — the <code>model</code> column is why the rows stay honest. |
| A sign-up asks for a payment card, or you do not have one                                                    | None of the three web chats should need one for this lab. If one asks, stop there.                                                 | Route P: ask the instructor to run your prompts on their account. Nobody needs a card or a subscription for this course.                                                             |
| The web chat will only sign you up with a phone number                                                       | Some services want a number where others take an email.                                                                            | That is a legitimate reason to choose route P. Do not give a number you would not give a stranger.                                                                                   |
| <code>ollama pull hf.co/... says file does not exist or 404</code>                                           | The tag after the colon must match a file in the repository exactly (Q4_K_M, upper case).                                          | Copy the name from <code>README.txt</code> ; check the Files tab on the model card for the spelling.                                                                                 |
| <code>cd ~/code/ITAS170</code> says no such directory, but <code>ls ~/code</code> shows <code>itas170</code> | You made the folder in lower case in week 1. Linux treats those as two different names, and every handout uses the upper-case one. | <code>mv ~/code/itas170 ~/code/ITAS170</code> — once. Git does not mind; your remote is unchanged.                                                                                   |

Today's one-sentence summary for `AGENTS.md`: **local for private and plentiful, frontier for hard and rare — and never trust a number you did not measure.**